

Name	IPs	Description	Actions
Accounts	192.168.1.100	Accounts Computer Systems	
AdminPCs	192.168.1.101	Internal administrator systems	
DMZServers	4.2.2.1, 8.8.8.8, 4.2.2.2	External DNS Servers	
EnggPCs	192.168.1.102	Computers used by Engineers	
ExternalFTP	192.168.1.103	External FTP Servers	
MailServers	74.125.88.100, 173.19.238.100	Servers required to access Gmail	
LBTPhone	203.129.224.90	IP Address for LBTP Phone Website	
Internal	192.168.1.0/24	Internal Systems	

Figure 3: IP aliases

Note: A default anti lockout rule is configured to ensure admin access to the firewall from the internal network. Take care not to disable this rule, otherwise you will be locked out of the firewall.

Diagnostics

Often, the implementer will stare at various inaccessible sites/services after such rules are implemented. The best way to diagnose these issues is to browse **Status – System Logs – Firewall**.

All 'blocked' log entries will be seen on this page, ordered chronologically. The button can be used to filter the results based on the source IP address, destination IP address, etc.

Missing rule?

Readers may revisit the ACL in Table 1 before reading

Firewall / Rules / Edit

Edit Firewall Rule

Action: ☐ Allow ☒ Deny

Choose what to do with packets that match the criteria specified below. Note: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

☒ **Disable this rule**
Set this option to disable this rule without removing it from the list.

Interface: LAN
Choose the interface from which packets must come to match this rule.

Address Family: IPv4
Select the Internet Protocol version this rule applies to.

Protocol: TCP/UDP
Choose which IP protocol this rule should match.

Source: ☐ Invert match ☐ Single host or alias ☐ Subnet/mask ☐ 192.168.1.0/24

Display Advanced: ☒ Show Advanced

Destination: ☐ Invert match ☐ Single host or alias ☐ Subnet/mask ☐ 203.129.224.90

Destination port range: From: 80 To: 80 (80)
Specify the destination port or port range for this rule. The 'To' field may be left empty if only starting a single port.

Extra Options:

Log: ☒ Log packets that are handled by this rule
After the firewall has finished local log space. Don't turn on logging for everything, if doing a lot of logging, consider using a remote logging server (see the System -> Setup -> Logging page).

Description: All Systems can access DNS Servers
A description may be entered here for administrative reference.

Advanced Options: ☒ Advanced Options

Advanced Options:

Source OS: Any
Note: this only works for TCP/UDP. General OS choice matches all subtypes.

Schedule: none
Leave as 'none' to leave the rule enabled all the time.

Gateway: ☐ Disabled: Don't send for this and other DNS connections.
Leave as 'default' to use the system routing table. ☐ Choose a gateway to utilize policy based routing.

Figure 4: DNS rules

the next sentence.

Once configuration is implemented according to the ACL defined in Table 1, LAN users will not be able to ping the Internet or even the firewall itself, since ICMP packets are not allowed towards the firewall and towards the Internet. The importance of ICMP packets for troubleshooting cannot be emphasised enough. Here, two rules will be required to allow ping requests towards pfSense and towards external DNS servers, at the minimum.

Note: To ensure correct documentation, update the ACL table first and then change the corresponding rule in the firewall.

An important configuration for accessing Gmail via the client

Readers will definitely notice that the *imap.gmail.com* and *smt.gmail.com* IP addresses keep changing practically every time. Due to this, it is difficult to create a rule to allow outgoing traffic towards these domain names.

About using FQDN for the host alias, the pfSense website has the following caveat: "DNS names that use very low TTLs and change frequently, such as round robin entries, are not reasonable to use in this fashion. This means that large sites like *google.com*, which return a different set of IP addresses with each query, would not be viable in an alias."

Thus, using various domain names such as *smt.gmail.com*, *imap.gmail.com*, etc, as alias entries will not work

Firewall / Rules / LAN

Floting: none LAN: (pfSense)

Rule	Protocol	Source	Port	Destination	Port	Gateway	Source Schedule	Description
210.05	any	*	*	LAN-Address	80	*	*	Anti-Lockout Rule
010.0	IPV4	LAN-IP	*	*	*	*	any	Default allow LAN to any rule
010.0	IPV4	LAN-IP	*	*	*	*	any	Default allow LAN-IP to any rule
010.0	any	Subnet/mask	*	DMZServers	80 (80)	Default	any	All Systems can access DNS Servers
010.05	IPV4 TCP	AdminPCs	*	GmailServerURL	GmailServer	Default	any	AdminPCs can access GmailServer using GmailServerURL
010.0	IPV4 TCP/UDP	Accounts	*	LBTPPhone	PhoneLBTPPhone	Default	any	Accounts can access LBTP Phone website using PhoneLBTP
010.05	IPV4 TCP/UDP	AdminPCs	*	InternetServices	Default	any	any	AdminPCs can access Internet services using InternetServices
010.0	IPV4 TCP	AdminPCs	*	MailServerURL	MailServer	Default	any	AdminPCs can access MailServer using MailServerURL
010.0	IPV4 TCP/UDP	EnggPCs	*	GmailServer	GmailServer	Default	any	EnggPCs can access GmailServer using GmailServer
010.0	IPV4 TCP/UDP	EnggPCs	*	ExternalMailServer	21 (80)	Default	any	EnggPCs can access External Mail Server using ExternalMailServer
010.0	IPV4	any	*	This Firewall	*	*	any	Internal PC's can ping Firewall
010.0	IPV4	any	*	DMZServers	*	Default	any	Internal PC's can ping DMZServers

Figure 5: LAN to WAN firewall rules